

Achievers

Foundation course for your success

Workbook - 4

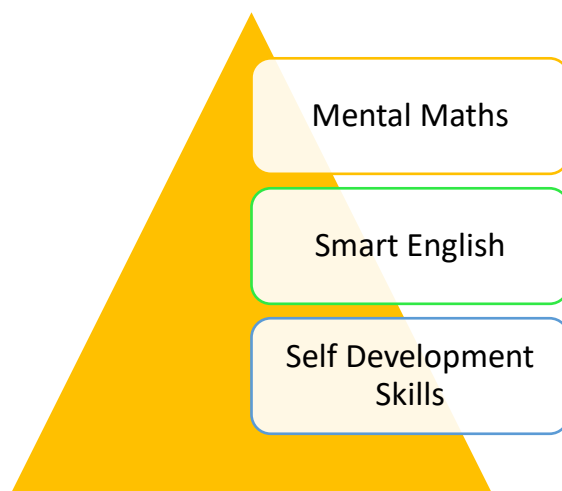
Class 7

Preface

Most students and teachers think that the word "**Worksheet**" is a synonym for the word "*Boring.*" We envision hours of sitting still at a desk-silent busy working for them and hours of grading for us. But worksheets still have a place in your classroom? In an age of open inquiry and problem-based learning, your child can still find power in a simple paper and pencil activity by practicing MENTORS BASKET WEEKLY worksheets.

Achievers Model bring to your child an opportunity to develop academically as well as for the competitive World. The Learning Process of every child is dependent on One's Ability to think logically and reason Quickly and effectively. These two factors are hence very crucial to a child's progress in education.

Mentors Basket Achievers weekly workbook covers Mental Math, Smart English, Enhance Self-Development Skills. These Powerful worksheets can help in building number sense, computational thinking, and problem-solving skills in your child at beginning of the foundational years. **Every Month we would be providing 3 Workbook and 1 Challenge Workbook, each of these are designed at different levels and standard wise. We would be providing upgraded workbooks, that makes the child to look forward to challenges. Weekly Workbook would be consisting of the following content:**



It is a unique way to provide differentiated brain-building work for different learners.

Strong Foundation Helps to Build Brighter Future

Design Framework

The Sample Workbook Consist of the topics as under:

- **Mental Maths** - We want our students to be able to understand content and concepts. We also want to stretch them to think and approach problems in new ways. This editable graphic organizer can help them do that! Challenge the students to solve a math problem more than one way. Have them explain how and why they solved it with their given strategy. Over time, we can see how mathematics can be transformed into a subject that is not only fun but also challenges students' creativity and independent thinking.
- **Smart English** - The study of English Language Arts, or speaking, listening, reading, and writing, is critical both to academic success and to life beyond school. Without the skills learned in language arts, students cannot achieve any kind of mastery in any of their coursework. Good communication skills are necessary for every career field, so these on-demand workbooks give your students the mixed review of grammar, spelling, and writing skills that will enable them to practice their language arts skills in a fun and diverting manner, ensuring that the skills they have learned will Supplement their language.
- **Self – Development** – Children who become invested in self-improvement at a young age will likely experience many advantages in life. But it can be a little tricky to figure out exactly how to go about teaching kids about self-improvement. We have developed strategies that can help children to be more confident and in becoming the best version of themselves.

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GUIDELINES

1. Look at Problems with Different Perspectives Through Observation, Exploration and Critical Thinking.
2. Integrate the Application of Math to Problem Solving.
3. Integrate the Reading Skill in English To Solve Vocabulary or Spelling Questions.
4. Think Critically and Reason Logically
5. Practise, Investigate, Evaluate and Learn with Understanding
6. Develop Positive Attitude Towards Finding Solution
7. If Your No Able to Solve or Find Solution Seek for Help.
8. Take A Break and Revisit the Question with Fresh Thought
9. Ask Yourself How Many Questions You Could Answer in First Attempt.
10. Focus and Concentration Is the Only Key to Success

Smart English Worksheet 1

1. Read the Passage and answer the questions.

Grasshoppers

Written and Photographed by Sandie Lee

It's a summer-time singer. Its body has several legs, wings, and eyes. Farmers either love it or hate it. What is it? A grasshopper of course.

There's between 11,000 and 18,000 species of grasshoppers worldwide and they're found everywhere except the North and South Poles. And even though they're common, they're also quite fascinating.

The Body

Grasshoppers have an exoskeleton [ek-soh-skel-i-tin] which means the skeleton is on the outside of their body. They have two sets of wings for flying – a front pair that is tough and rigid and a hind pair that is soft and flexible. The grasshopper also has three pairs of very strong legs. They use these to walk and to jump 20 times their own body length. Imagine if you could do that!

Have you ever tried to catch a grasshopper? Chances are it bounced away before you had a chance. That's because this insect has five eyes. Two eyes are on the front of the head. These are made up of thousands of single lenses that can see forward, backward, and sideways. Two more eyes are located on the end of each antenna. The final eye is between the antennae. It's no wonder they're so quick!

Grasshoppers don't have ears, but that doesn't mean they're deaf. They actually hear through their knees. The tympanum [tim-puh-nuh-m], a drum-like organ, can pick up vibrations from the air and on the ground.



Music Makers

If you hear a chirp chirp chirp on a summer's night, it could be a male grasshopper trying to attract a mate, or warn off other males. They do this by rubbing their back leg against their forewing - A row of pegs along the inside of the grasshopper's leg acts like a file. When this is rubbed on the wing it causes a vibration which produces the sound.

Other grasshoppers like the **Band-Winged Grasshopper** will snap their hind wings rapidly when in flight. This makes a weird crackling sound.

Friend or Foe?

Grasshoppers like to eat most plants, but some favor the flavor of wheat, oats, corn, barley, rye, clover, alfalfa, and cotton – all things we use and farmers hate to lose. In fact, a large group of grasshoppers called locusts can quickly and easily devour a farmer's crop in a matter of minutes.

However, some grasshoppers are actually a help to farmers. The **Turnbull** is a type of grasshopper that will dine on the weeds that kill crops. Other grasshoppers like the **Two-Striped** grasshopper will eat plants that are toxic to cattle.

Now that you know some interesting facts about grasshoppers, be sure to be on the look out for this summer-time favorite. You can learn more about the great grasshopper by visiting your local library or the internet.

Fascinating Grasshopper Facts

- The smallest grasshopper is the **Pygmy Grasshopper**. It's only 20 mm long!
- The largest grasshopper is called the **Giant Grasshopper** and measures in at 60-90 mm for females and 45-55mm for males.
- If you grab a grasshopper you may get "spit" on. This strong, brown, gooey, liquid is called tobacco juice and is used to deter predators.



Smart English Worksheet 1

Name: _____

Grasshoppers

Written and Photographed by Sandie Lee



1. Circle the best answers to these questions.

- a. How many wings does a grasshopper have? 2 4 6 8 10
- b. How many eyes does a grasshopper have? 1 3 5 7 9
- c. About how long are the world's largest grasshoppers? 20mm 55mm 90mm

2. Grasshoppers do not have ears, yet they can hear. Explain.

3. How does a grasshopper make a chirping sound?

- a. It rubs two legs together.
- b. It rubs one of its legs along one of its wings.
- c. It rubs its wings together.
- d. It rubs its back leg against one of its antennae.

4. Why do farmers sometimes dislike grasshoppers?

5. Why do farmers sometimes like grasshoppers?

Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
5			

Smart English Worksheet 2

Name: _____

Grasshoppers Vocabulary Activity



Step 1: Re-read the article, "Grasshoppers." As you read, use a highlighter or yellow crayon to highlight each of the vocabulary words from the box.

exoskeleton	organ	vibration
foe	barley	toxic

Step 2: Match each vocabulary word with its definition.

- | | |
|----------------------|--|
| 1. _____ exoskeleton | a. enemy |
| 2. _____ organ | b. poisonous; dangerous to eat or drink |
| 3. _____ vibration | c. movement of molecules that makes a sound |
| 4. _____ foe | d. body part made of cell tissue |
| 5. _____ barley | e. hard covering on the outside of an animal's body |
| 6. _____ toxic | f. type of grain used to make cereal and other food products |

Step 3: Complete each sentences with the correct vocabulary word.

7. You shouldn't eat those berries because they might be _____.
8. After the storm, the farmer's _____ field was flooded.
9. When his brother turned up the music in the car, Joey could feel _____ in the seat.
10. The liver is the heaviest _____ in the human body.

Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
10			

Smart English Worksheet 3

Pronouns: I or Me?

I and **me** are first person pronouns.

I is used as a subject:

examples: **I** wanted to go to the park on Saturday.
Libby and **I** are selling candy bars for our volleyball team.

Me is used as an object.

examples: The present with the big red bow is for Zoe and **me**.
Choose **me** for class treasurer.
Dad told Ian and **me** a funny joke.

Circle the pronoun that correctly completes each sentence.

1. This song has been difficult for (I / me) to learn.
2. It's a song that my sister and (I / me) both enjoy.
3. My sister, Ava, and (I / me) are practicing the song every day.
4. Dad said he would pay my brother and (I / me) ten dollars to clean the garage.
5. I asked if he and (I / me) could earn more money by having a yard sale.
6. Dad thanked my brother and (I / me) for doing a good job.
7. Jacob and (I / me) made a birthday cake for our little sister, Mia.
8. Jacob baked the cake, and (I / me) frosted and decorated it.
9. Mia thanked Jacob and (I / me) for the best chocolate cake she'd ever tasted.
10. Kaylee and (I / me) decided to wash our dog, Buster, in the bathtub.
11. Before we could dry him off, Buster shook water all over her and (I / me).
12. "Look at all this dog hair in the tub!" Kaylee and (I / me) shouted together.
13. Alexis sat beside Chloe and (I / me) at the picnic table.
14. Alexis had grapes, but (I / me) had apple slices.
15. Alexis gave Chloe and (I / me) some of her grapes.

Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
15			

Smart English Worksheet 4

- Creative Writing:** Read the instructions carefully and explore your imagination to write the most appropriate information.



Your Score

Total Marks	Correct Answer	Wrong Answer	Not Attempted Questions
10			

Smart English Worksheet 5

1. **Synonyms Counts!** In this exercise you will come across new words, add on to your knowledge by finding the meaning. Complete the task as instructed.

What Does It Mean?

Using Context Clues

Read each sentence below. Use the other words in the sentence to help you figure out the meaning of the underlined word. Fill in the circle for the correct answer.

1. He saw in the distance a beautiful, graceful bird soaring across the sky.

☐ falling ☐ gliding ☐ leaping

2. The little boy became drowsy as the day turned into night.

☐ desperate ☐ active ☐ tired

3. He was amazed at the beauty he saw above him.

☐ pleased ☐ disappointed ☐ astonished

4. Little sparrow began to feel as powerful as the giant bird.

☐ large ☐ strong ☐ beautiful

5. When the boy became chief, he ruled with wisdom.

☐ many riches ☐ no help ☐ understanding

6. The people watched with awe as the eagle flew above them.

☐ anger ☐ terror ☐ wonder

7. The pterodactyl was gliding high above the ground.

☐ flapping ☐ moving smoothly ☐ spinning

8. The boy was very confused after he bumped his head.

☐ mixed up ☐ relaxed ☐ excited

9. The castle looked enormous in the background.

☐ very big ☐ far ☐ near

Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
9			

Smart English Worksheet 6

1. Read the instruction carefully and make sentences, an example has been done for you.

Prepositional Phrases As Adjectives

A **prepositional phrase** is a group of words beginning with a preposition and ending with a noun or pronoun. The noun or pronoun at the end of the phrase is called the object of the preposition. Often there are other descriptive words between the preposition and its object.

Sometimes prepositional phrases do the job of an adjective. The prepositional phrase tells us which one.

example: The little boy in the red jacket is my brother, Henry.

In the red jacket is the prepositional phrase. In is the preposition, and jacket is the object. Red describes the jacket. The entire phrase, in the red jacket, tells us which boy.



Underline the prepositional phrase in each sentence below. Circle the word it describes.

1. My favorite place to walk is the park across the street.
2. The wooden bridge over the muddy creek was built last year.
3. Those trees beside the lake are willows.
4. The tall evergreen trees behind the playground are spruce trees.
5. The robin in the nest is watching her young.
6. The five frogs on the lily pads are catching dragonflies.
7. The dragonflies with shimmering wings zip away.
8. The squirrels with the bushy tails ran past me.
9. The grass beneath the swings is turning brown.
10. The park near my house is a special place.

❖ **Now try this:** Write your own sentence that has a prepositional phrase used as an adjective. Underline the prepositional phrase. Circle the word it describes.

Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
10			

Mental Maths Worksheet 1

1. Find the Solution:

1.1 For every 10 used electric motors turns into Peter's recycling shop. Peter gets paid Rs 550. It costs Peter Rs 150 to dispose of each motor. What is the net income Peter generating from recycling 100 used electric motors?	Solution 1.1
1.2 Nidhi planted 10 strawberry plants. For four weeks in a row, each plant produced 5 strawberries. Nidhi shared her strawberries equally among her 4 best friends and herself. How many strawberries did each person get?	Solution 1.2
1.3 Mrs Sharma bought 13 new hats for her son's team. The cost of the hats was Rs 80 each. How much did Mrs Sharma pay for these 13 new hats. Write the equation and solve it.	Solution 1.3

Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
3 x 5 marks			

Mental Maths Worksheet 2

1. Find the solution by calculating it visually. Can you do it in 5 Min?

1. Each side of a regular pentagon is 48.2 cms.What is the perimeter	
2. The circle graph has four sections.Only three sections are labelled.The labels are 13.36%,15.55%,24.09%.What should the missing section be?	
3. The angles in a quadrilateral measure 110,91,115, F degrees.What is the value of F	

Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
3 x 4 marks			

Mental Maths Worksheet 3

1. Let us try something different than the routine sums. Solve the following and quiz your brains.

Percentages

Part 1: Convert each percent into a fraction in simplest form.

a. 25% - _____

b. 4% - _____

c. 68% - _____

b. 20% - _____

d. 50% - _____

e. 7% - _____

Part 2: Convert each fraction into a percent.

f. $\frac{2}{5}$ - _____

g. $\frac{13}{25}$ - _____

h. $\frac{7}{25}$ - _____

i. $\frac{19}{100}$ - _____

j. $\frac{3}{4}$ - _____

k. $\frac{19}{20}$ - _____

Part 3: Find the answer to the word problem.

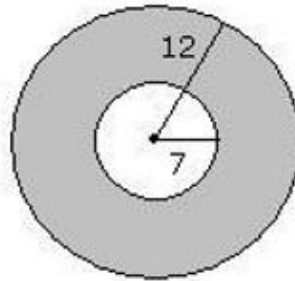
- l. Maria took a history test with 25 questions on it. She correctly answered 22 questions. Write her test score as a _____

Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
13			

Mental Maths Worksheet 4

1. In the figure given below, a circle is inscribed inside another circle. The radius of the outer circle is 12 cm and that of the inner circle is 7 cm. Find the area of the shaded portion between the circles.



2.

Use the sign of $>$, $<$ or $=$ in the box to make the statements true:

- (a) $(-8) + (-4) \square (-8) - (-4)$
 (b) $(-3) + 7 - (19) \square 15 - 8 + (-9)$
 (c) $23 - 41 + 11 \square 23 - 41 - 11$
 (d) $39 + (-24) - (15) \square 36 + (-52) - (-36)$
 (e) $(-231) + 79 + 51 \square (-399) + 159 + 81$

3. Given the line of symmetry, find the other hole(s) in the following figures.

i)		
ii)		

Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
8			

Mental Maths Worksheet 5

Probability



1. If you roll a die, what are the chances of rolling a two? answer: _____

2. If you roll a die, what is the probability that you will roll an even number? answer: _____

3. A bag contains 3 red marbles, 3 blue marbles, and 1 green marble. If a marble is drawn from the bag at random, what is the probability that the marble will be blue? answer: _____

4. A bag contains 6 number tiles. The numbers in the bag are 3, 7, 8, 9, 13, and 15. If you randomly draw one tile from the bag, what is the probability of picking an odd number? answer: _____

5. Mr. Jones has a hot air balloon. Because the basket is so small, he can take one child for a ride with him. Mary, Carla, John, Lynda, Peter, and Janessa all want to go. They each write their name on a piece of paper and place them in a hat. Mr. Jones randomly selects one child to go with him.

What is the probability that he will select a boy? answer: _____

What is the probability that he will select a girl? answer: _____

6. John and Jackie are rolling a die. John wins if he rolls a number higher than 4. Jackie wins if the number rolled is 4 or less. Is this game fair? Explain.

Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
6			

Mental Maths Worksheet 6

Congruent Shapes

CONGRUENT figures are the same size and shape.



CONGRUENT



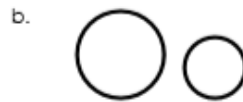
CONGRUENT



NOT CONGRUENT

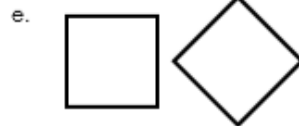
Write CONGRUENT or NOT CONGRUENT for each.





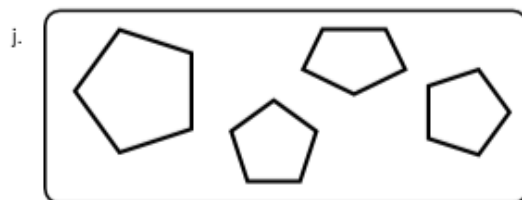
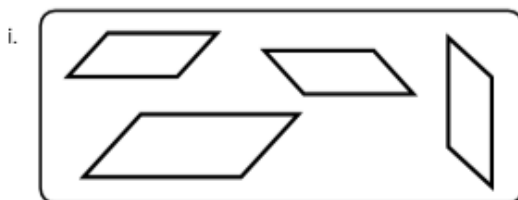
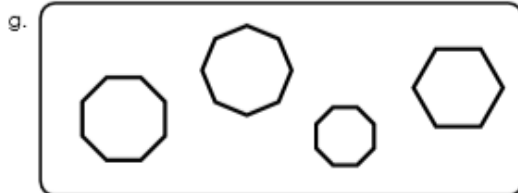








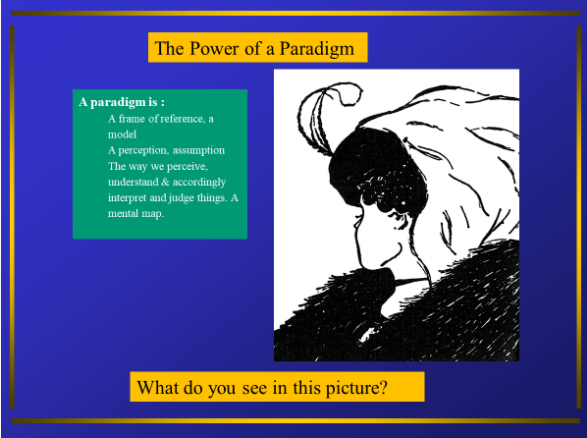
Color all the congruent shapes in each box.



Your Score

Total Questions	Correct Answer	Wrong Answer	Not Attempted Questions
10			

My Journal – Identifying Myself

 The Power of a Paradigm A paradigm is : A frame of reference, a model A perception, assumption The way we perceive, understand & accordingly interpret and judge things. A mental map. What do you see in this picture?	Be → See → Think → Feel → Behave We cannot change our seeing without changing our being Our habits paradigm needs to be redefined: An “inside-out” approach to personal and interpersonal effectiveness
Pls explain what you see 2 types of people in above picture?	
Name your 5 positive habits?	
Since it's Father's Day this month Name 2 habits that you admire in your dad?	